Practical 9
Programs on Files and Streams

Program 1: Create and Write to a File

Aim : Create a text file and write student information into it.

Program

```
using System;
using System.IO;

namespace FilesStreamsDemo
{
   class Program
   {
      static void Main(string[] args)
      {
         string fileName = "student.txt";

         File.WriteAllText(fileName,
            "Name: Rahul\nCourse: BSc IT\nRoll No: 101");

         Console.WriteLine("Data written successfully.");
         Console.WriteLine("File Name: " + fileName);

         Console.ReadKey();
      }
   }
}
```

Output

```
Data written successfully.
File Name: student.txt
```

Program 2: Read Data from a File

Aim

Read and display the contents of a text file.

Program

```
using System;
using System.IO;

namespace FilesStreamsDemo
{
```

```
    class Program
    {
        static void Main(string[] args)
        {
            string fileName = "student.txt";

            if (File.Exists(fileName))
            {
                string data = File.ReadAllText(fileName);

                Console.WriteLine("File Contents:");
                Console.WriteLine(data);
            }
            else
            {
                Console.WriteLine("File does not exist.");
            }

            Console.ReadKey();
        }
    }
}
```

Output

```
File Contents:
Name: Rahul
Course: BSc IT
Roll No: 101
```

Program 3: Append Data to a File

Aim

Add new data to an existing file without deleting its previous contents.

Program

```
using System;
using System.IO;

namespace FilesStreamsDemo
{
    class Program
    {
        static void Main(string[] args)
        {
            string fileName = "student.txt";

            File.AppendAllText(fileName,
                "\nMarks: 85");
```

```
            Console.WriteLine("Data appended successfully.");

            Console.ReadKey();
        }
    }
}
```

Program 4: Copy, Move and Delete a File

Aim

Perform basic file-system operations.

Program

```
using System;
using System.IO;

namespace FilesStreamsDemo
{
    class Program
    {
        static void Main(string[] args)
        {
            string original = "student.txt";
            string copy = "student_backup.txt";
            string moved = "student_new.txt";

            // Create original file
            File.WriteAllText(original, "Student information");

            // Copy file
            File.Copy(original, copy, true);
            Console.WriteLine("File copied.");

            // Move file
            File.Move(copy, moved);
            Console.WriteLine("File moved.");

            // Delete moved file
            File.Delete(moved);
            Console.WriteLine("File deleted.");

            Console.ReadKey();
        }
    }
}
```

Output

```
File copied.
File moved.
```

File deleted.

Program 5: Create and Manage a Directory

Aim

Create a folder and store a file inside it.

Program

```
using System;
using System.IO;

namespace FilesStreamsDemo
{
   class Program
   {
      static void Main(string[] args)
      {
         string folderName = "StudentData";

         if (!Directory.Exists(folderName))
         {
            Directory.CreateDirectory(folderName);
            Console.WriteLine("Directory created.");
         }

         string fileName =
            Path.Combine(folderName, "student.txt");

         File.WriteAllText(fileName,
            "Name: Priya\nCourse: BSc IT");

         Console.WriteLine("File created inside directory.");

         Console.ReadKey();
      }
   }
}
```

Output

```
Directory created.
File created inside directory.
```

Program 6: Display All Files in a Directory

Aim

Display all files present in a specified directory.

Program

```
using System;
using System.IO;

namespace FilesStreamsDemo
{
   class Program
   {
      static void Main(string[] args)
      {
         string folder = "StudentData";

         if (Directory.Exists(folder))
         {
            string[] files = Directory.GetFiles(folder);

            Console.WriteLine("Files in directory:");

            foreach (string file in files)
            {
               Console.WriteLine(file);
            }
         }
         else
         {
            Console.WriteLine("Directory does not exist.");
         }

         Console.ReadKey();
      }
   }
}
```

Explanation

Directory.GetFiles() returns the names of files present in the specified directory.

Program 7: Write Data Using StreamWriter

Aim

Write multiple lines to a file using StreamWriter.

Program

```
using System;
using System.IO;

namespace FilesStreamsDemo
{
   class Program
   {
      static void Main(string[] args)
```

```
        {
            using (StreamWriter writer =
                  new StreamWriter("students.txt"))
            {
                writer.WriteLine("Student Details");
                writer.WriteLine("----------------");
                writer.WriteLine("101 - Rahul");
                writer.WriteLine("102 - Priya");
                writer.WriteLine("103 - Amit");
            }

            Console.WriteLine("Data written using StreamWriter.");

            Console.ReadKey();
        }
    }
}
```

Output

Data written using StreamWriter.

The file students.txt will contain:

```
Student Details
----------------
101 - Rahul
102 - Priya
103 - Amit
```

Program 8: Read Data Using StreamReader

Aim

Read a text file line by line using StreamReader.

Program

```
using System;
using System.IO;

namespace FilesStreamsDemo
{
    class Program
    {
        static void Main(string[] args)
        {
            using (StreamReader reader =
                  new StreamReader("students.txt"))
            {
                string line;
```

```
            while ((line = reader.ReadLine()) != null)
            {
                Console.WriteLine(line);
            }
        }

        Console.ReadKey();
    }
  }
}
```

Explanation

ReadLine() reads one line at a time. The loop continues until the end of the file is reached.

Program 9: Copy a File Using FileStream

Aim

Copy the contents of one file to another using FileStream.

Program

```
using System;
using System.IO;

namespace FilesStreamsDemo
{
  class Program
  {
    static void Main(string[] args)
    {
      string source = "students.txt";
      string destination = "students_copy.txt";

      using (FileStream input =
          new FileStream(source, FileMode.Open))
      using (FileStream output =
          new FileStream(destination, FileMode.Create))
      {
        input.CopyTo(output);
      }

      Console.WriteLine("File copied using FileStream.");

      Console.ReadKey();
    }
  }
}
```

Program 10: Write and Read Bytes Using FileStream

Aim

Write data as bytes and read it using FileStream.

Program

```
using System;
using System.IO;
using System.Text;

namespace FilesStreamsDemo
{
   class Program
   {
      static void Main(string[] args)
      {
         string fileName = "data.txt";

         string text = "Hello from FileStream!";

         byte[] data = Encoding.UTF8.GetBytes(text);

         // Writing
         using (FileStream stream =
               new FileStream(fileName, FileMode.Create))
         {
            stream.Write(data, 0, data.Length);
         }

         // Reading
         using (FileStream stream =
               new FileStream(fileName, FileMode.Open))
         {
            byte[] readData = new byte[stream.Length];

            stream.Read(readData, 0, readData.Length);

            string result =
               Encoding.UTF8.GetString(readData);

            Console.WriteLine(result);
         }

         Console.ReadKey();
      }
   }
}
```

Output

Hello from FileStream!